103 Temperature Changes and Heat Capacity

1.3.1 Power: the rate of energy transfer

Power measures energy transferred or converted per unit time. Average power is the total energy transferred divided by the elapsed time; 1 W means 1 J per second. Transferring the same energy in less time requires greater average power. If power varies, this average need not equal the instantaneous power at any chosen moment.

$$P = \frac{Q}{t}$$

P: average power (W); Q: energy transferred (J); t: time interval (s).

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1.3.2 · 1.3.3 Heat capacity and temperature change

Heat capacity is the energy needed to raise the temperature of a whole object by 1 K, equivalent to a rise of 1 °C. With no change of state and approximately constant heat capacity, absorbed energy equals heat capacity times temperature rise. A larger heat capacity gives a smaller rise for the same energy input; it depends on both material and mass.

$$Q = C\Delta T$$

$$C = \frac{Q}{\Delta T}$$

C: heat capacity (J K^{-1}); Q: heat supplied (J); ΔT: temperature rise (K or °C). No change of state.

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1.3.4 · 1.3.5 Specific heat capacity and heat capacity

Specific heat capacity is the energy needed to raise 1 kg of a substance by 1 K. The heat capacity of a whole object is its mass multiplied by its specific heat capacity. Doubling the mass of the same material doubles heat capacity but does not change specific heat capacity; the two quantities also have different units.

$$c = \frac{C}{m}$$

c: specific heat capacity (J kg^{-1} K^{-1}); C: heat capacity (J K^{-1}); m: mass (kg).

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1.3.6 Using the specific heat equation

With no change of state and approximately constant specific heat capacity, the absorbed energy is mass times specific heat capacity times temperature rise. For cooling, either calculate released energy using the magnitude of the temperature drop or consistently use signed changes. If energy escapes to the surroundings, not all the heater's input is absorbed by the sample.

$$Q = mc\Delta T$$

$$c = \frac{Q}{m\Delta T}$$

Q: heat supplied (J); m: mass (kg); c: specific heat capacity (J kg^{-1} K^{-1}); ΔT: temperature rise (K or °C). Constant c; no change of state.

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1.3.7 Energy balance in heat exchange

Treat the hot body, cold body and container as one system. With no energy supplied from outside or lost to the surroundings, energy released by the hotter parts equals energy absorbed by the other parts. Calculate each temperature change and any change of state separately; include the container's heating and any latent heat required.

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1.3.8 Conservation of energy

Energy can be transferred between objects and converted between kinetic, potential, internal and other forms, but cannot be created or destroyed. Total energy remains constant in an isolated system. For a system exchanging energy with its surroundings, include all inputs and outputs. Friction reduces mechanical energy without violating conservation of total energy.

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